

CSTE 5131

Machine Learning

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Classification: Basic Concepts

- Classification problem overview
- General issues of classification DM
- Mining classification model by decision tree induction
- Bayesian Classification
- Other classification methods
- Summary

Decision Tree Example

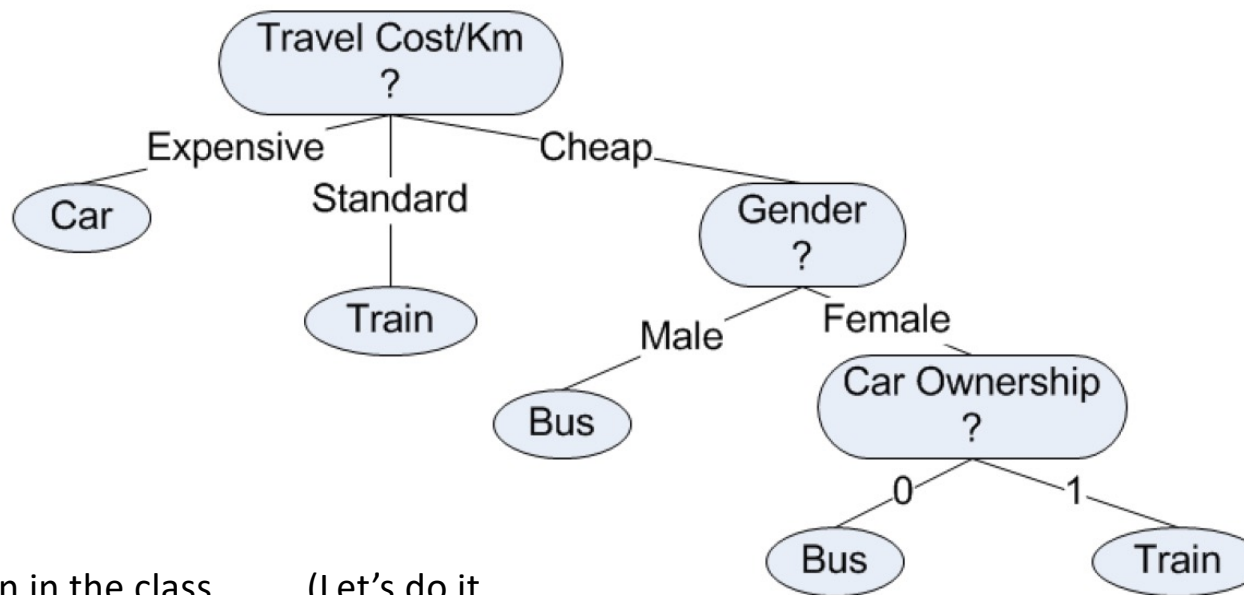
Training Data:

Attributes				Classes
Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Let's call this data as Table D

Decision Tree Example

- Based on above training data, we can induce a decision tree as the following:



Calculation in the class.....(Let's do it together!)



How to use the above decision tree?

- Decision tree can be used to predict a pattern or to classify the class of a data.
- Suppose we have new unseen records of a person from the same location where the data sample was taken.

Test Data:

Person name	Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation Mode
Alex	Male	1	Standard	High	?
Buddy	Male	0	Cheap	Medium	?
Cherry	Female	1	Cheap	High	?

How to use the above decision tree?

- The rules generated from the decision tree above are mutually exclusive and exhaustive for each class label on the leaf node of the tree:
 - **Rule 1:** If Travel cost/km is expensive then mode = car
 - **Rule 2:** If Travel cost/km is standard then mode = train
 - **Rule 3:** If Travel cost/km is cheap and gender is male then mode = bus
 - **Rule 4:** If Travel cost/km is cheap and gender is female and she owns no car then mode = bus
 - **Rule 5:** If Travel cost/km is cheap and gender is female and she owns 1 car then mode = train



How to use the above decision tree?

- Based on the rules or decision tree above, the classification is very straightforward.

Person name	Travel Cost (\$)/km	Gender	Car ownership	Transportation Mode
Alex	Standard	Male	1	Train
Buddy	Cheap	Male	0	Bus
Cherry	Cheap	Female	1	Train

How to generate a decision tree?

- How to measure impurity?

- Entropy
- Gini index
- Classification error

$$Entropy = \sum_i -p_j \log_2 p_j$$

Probability P_j of
a class j

$$Gini\ Index = 1 - \sum_j p_j^2$$

$$Classification\ Error = 1 - \max\{p_j\}$$

How to generate a decision tree?

- Compute probability of each class (based on the training data):

- $\text{Prob}(\text{Bus}) = 4/10 = 0.4$

- $\text{Prob}(\text{Car}) = 3/10 = 0.3$

- $\text{Prob}(\text{Train}) = 3/10 = 0.3$

- Entropy of the parent table D:

$$\text{Entropy} = - 0.4 \log_2(0.4) - 0.3 \log_2(0.3) - 0.3 \log_2(0.3) = 1.571$$

Note: Entropy of a pure table (consist of single class) is zero because the probability is 1 and $\log(1) = 0$.

How to generate a decision tree?

- From table D, we take out each attribute and its associate classes. If we have m attributes, then we will take out m subset of D. Let us call these subsets as S_i . Table D is the parent of table S_i
- Compute entropy (degree of impurity) for each subset
 - separate the computation of impurity degree for each value of the attribute i .

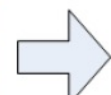
Travel Cost (\$)/km	Transportation mode
Cheap	Bus
Cheap	Bus
Cheap	Bus
Cheap	Bus
Cheap	Train
Expensive	Car
Expensive	Car
Expensive	Car
Standard	Train
Standard	Train



Travel Cost (\$)/km	Classes
Cheap	Bus
Cheap	Bus
Cheap	Bus
Cheap	Bus
Cheap	Train

4B, 1T

Entropy	0.722
Gini index	0.320
classification error	0.200



Travel Cost (\$)/km	Classes
Expensive	Car
Expensive	Car
Expensive	Car

3C

Entropy	0.000
Gini index	0.000
classification error	0.000



Travel Cost (\$)/km	Classes
Standard	Train
Standard	Train

2T

Entropy	0.000
Gini index	0.000
classification error	0.000

How to generate a decision tree?

- Compute information gain for each attribute
 - The measure to compare the difference of impurity degrees is called information gain
 - Information gain is computed as impurity degrees of the parent table and weighted summation of impurity degrees of the subset table

Information gain (i) = Entropy of parent table D – Sum (n_k/n * Entropy of each value k of subset table S_i)

n_k = number of records for each attribute values

n = total number of records

How to generate a decision tree?

Information gain (travel cost) = $1.571 - (5/10 * 0.722 + 2/10 * 0 + 3/10 * 0) = 1.210$

Information gain (gender) = .125

Information gain (car ownership) = .534

Information gain (income level) = .695

Optimum attribute -> attribute that produces the maximum information gain

- In this case, travel cost per km produces the maximum information gain

How to generate a decision tree?

- Our decision tree now consists of a single root node

Travel Cost/Km
?

- Once we obtain the optimum attribute, we can split the data table according to that optimum attribute. In our example, we split the data table based on the value of travel cost per km

Data

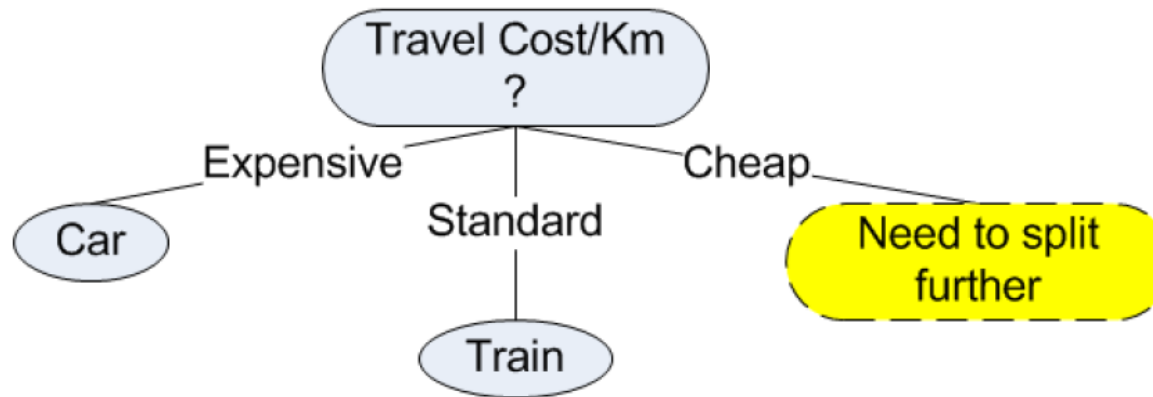
Attributes				Classes
Gender	Car	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Attributes				Classes
Gender	Car ownership	Travel Cost /km	Income Level	Transportation mode
Female	0	Cheap	Low	Bus
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train

Attributes				Classes
Gender	Car ownership	Travel Cost /km	Income Level	Transportation mode
Female	1	Expensive	High	Car
Female	2	Expensive	High	Car
Male	2	Expensive	Medium	Car

Attributes				Classes
Gender	Car ownership	Travel Cost /km	Income Level	Transportation mode
Female	1	Standard	Medium	Train
Male	0	Standard	Medium	Train

How to generate a decision tree?



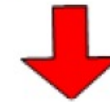
- For Cheap travel cost/km, the classes are not pure, thus we need to split further in the next iteration.

How to generate a decision tree?

- Second iteration:

For second iteration, our data table D only comes from the Cheap Travel cost/km. We remove attribute travel cost/km from the data because they are equal and redundant.

Attributes				Classes
Gender	Car ownership	Travel Cost /km	Income Level	Transportation mode
Female	0	Cheap	Low	Bus
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train



Data

Attributes			Classes
Gender	Car ownership	Income Level	Transportation mode
Female	0	Low	Bus
Male	0	Low	Bus
Male	1	Medium	Bus
Male	1	Medium	Bus
Female	1	Medium	Train

How to generate a decision tree?

- Compute entropy of the updated data table
- Repeat the procedure of computing degree of impurity and information gain for the three attributes.

Information gain (gender) = ?

Information gain (car ownership) = ?

Information gain (income level) = ?

- The maximum gain is obtained for the optimum attribute Gender

Attributes			Classes
Gender	Car ownership	Income Level	Transportation mode
Female	0	Low	Bus
Female	1	Medium	Train
Male	0	Low	Bus
Male	1	Medium	Bus
Male	1	Medium	Bus



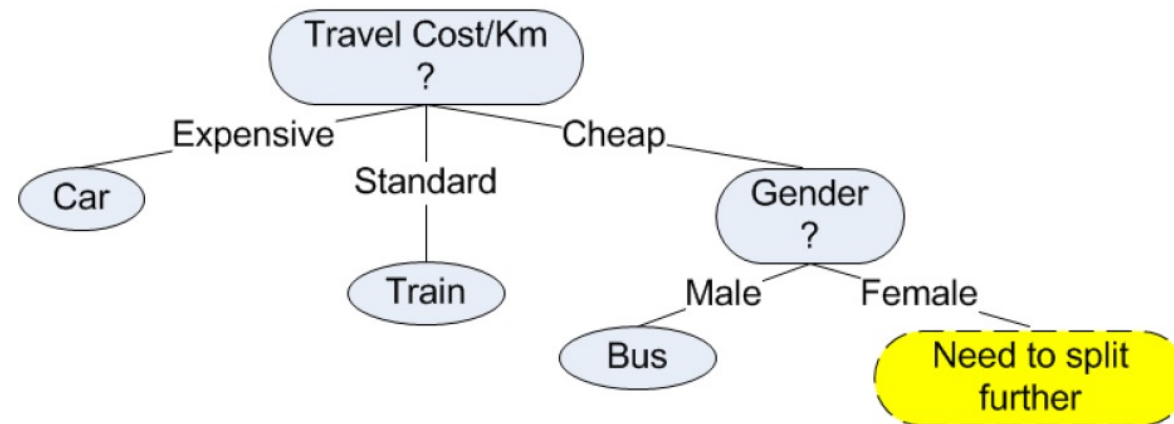
Attributes			Classes
Gender	Car ownership	Income Level	Transportation mode
Female	0	Low	Bus
Female	1	Medium	Train



Attributes			Classes
Gender	Car ownership	Income Level	Transportation mode
Male	0	Low	Bus
Male	1	Medium	Bus
Male	1	Medium	Bus



How to generate a decision tree?



How to generate a decision tree?

- Third iteration:

Data table of the third iteration comes only from part of the data table of the second iteration with male gender removed (thus only female part).

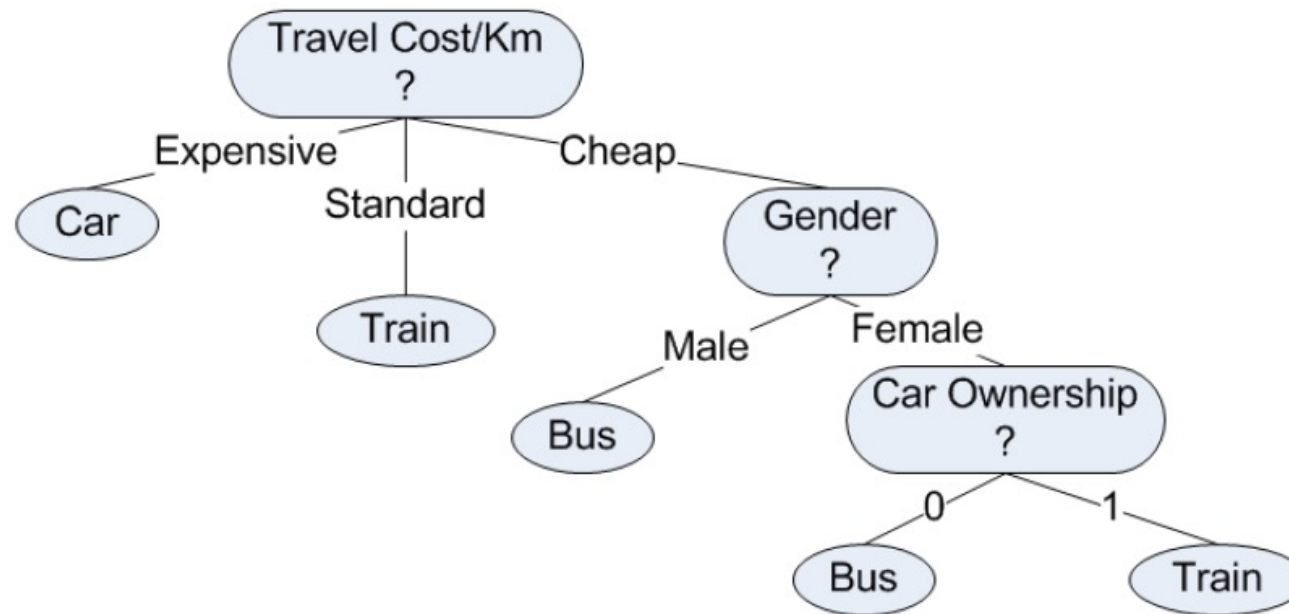
Attributes			Classes
Gender	Car ownership	Income Level	Transportation mode
Female	0	Low	Bus
Female	1	Medium	Train



Data third iteration

Attributes		Classes
Car ownership	Income Level	Transportation mode
0	Low	Bus
1	Medium	Train

How to generate a decision tree?



Homework!



- Read class note, and chapter 8 (mainly Section 8.1 and 8.2) of book “Data Mining Concepts and Techniques”

Thank you

